

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

09

10 questions · ~15 min

Q1

GRID-IN

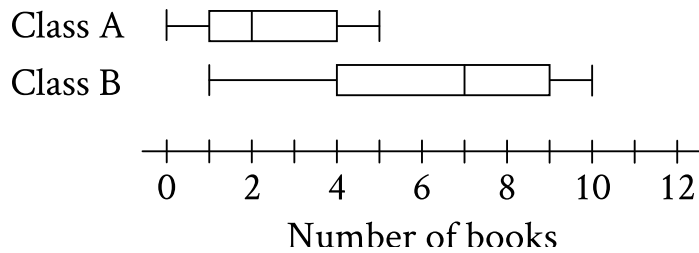
PROBLEM-SOLVING AND DATA ANALYSIS

A gift shop buys souvenirs at a wholesale price of 7.00 dollars each and resells them each at a retail price that is 290% of the wholesale price. At the end of the season, any remaining souvenirs are marked at a discounted price that is 80% off the retail price. What is the discounted price of each remaining souvenir, in dollars?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- From left to right, the values of the vertical bars in the Class A box plot are as follows:
 - First vertical bar: 0
 - Second vertical bar: 1
 - Third vertical bar: 2
 - Fourth vertical bar: 4
 - Fifth vertical bar: 5
- From left to right, the values of the vertical bars in the Class B box plot are as follows:
 - First vertical bar: 1
 - Second vertical bar: 4
 - Third vertical bar: 7
 - Fourth vertical bar: 9
 - Fifth vertical bar: 10

The two box plots show the distribution of number of books read over the summer by the students in two different English classes. What is the positive difference between the ranges of number of books read over the summer for the two classes?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2

3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN

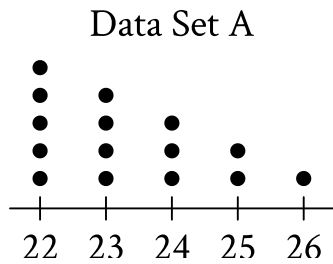
PROBLEM-SOLVING AND DATA ANALYSIS

The number a is 55% less than the number b . The number b is 320% greater than 160. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



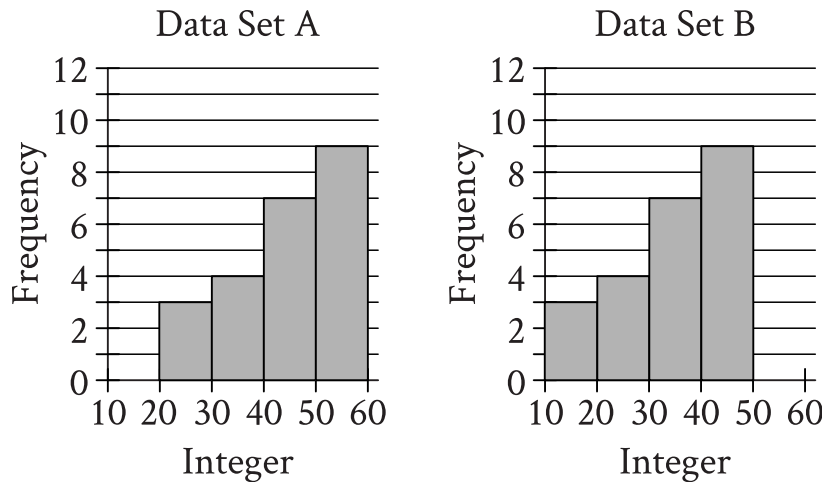
- The data for the dot plot are as follows:
 - 22: 5 dots
 - 23: 4 dots
 - 24: 3 dots
 - 25: 2 dots
 - 26: 1 dot

The dot plot represents the 15 values in data set A. Data set B is created by adding 56 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

- A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.
- B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

A scientist studying the life cycle of dragonflies counted the number of dragonflies in a certain habitat each day for 46 days. On February 15, there were 99 dragonflies in the habitat. The percent increase in the number of dragonflies in the habitat from January 1 to February 15 was 12.50%. How many dragonflies were in the habitat on January 1?

- A. 88
- B. 87
- C. 12
- D. 8



- Data Set A:
 - The horizontal axis is labeled Integer. It ranges from 10 to 60 and is divided into 5 equal intervals.
 - The vertical axis is labeled Frequency. It ranges from 0 to 12 in increments of 1, with values marked every 2 grid lines.
 - The histogram has a skewed right shape.
 - The histogram has 4 bins.
 - The Frequency data for the 4 bins are as follows:
 - 20 to 30: 3
 - 30 to 40: 4
 - 40 to 50: 7
 - 50 to 60: 9
- Data Set B:
 - The horizontal axis is labeled Integer. It ranges from 10 to 60 and is divided into 5 equal intervals.
 - The vertical axis is labeled Frequency. It ranges from 0 to 12 in increments of 1, with values marked every 2 grid lines.
 - The histogram has a skewed right shape.
 - The histogram has 4 bins.
 - The Frequency data for the 4 bins are as follows:
 - 10 to 20: 3
 - 20 to 30: 4

- 30 to 40: 7
- 40 to 50: 9

Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- A. 0
- B. 1
- C. 10
- D. 23

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

After 20% of the original number of marbles in a group were removed from the group, 360 marbles remained in the group. How many marbles were removed from the group?

- A. 72
- B. 90
- C. 450
- D. 1,800

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

- A. Median
- B. Mean
- C. Sum of the numbers
- D. Standard deviation

Q9

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

210 is $p\%$ greater than 30. What is the value of p ?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The table shown summarizes the number of employees at each of the 17 restaurants in a town.

Number of employees	Number of restaurants
2 to 7	2
8 to 13	4
14 to 19	2
20 to 25	7
26 to 31	2

Which of the following could be the median number of employees for the restaurants in this town?

- A. 2
- B. 9
- C. 15
- D. 21